



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit V / Peephole Optimization & Optimization of Basic Blocks**

- **Course Outcome: CO5**

- **Topic Learning Outcome: TLO13**

- **Activity Chosen: Minute Paper**

- **Justification:**

The concept of optimization is important for the design of optimizing compiler. The teacher must determine the students' level of knowledge on this topic. Therefore, the one-minute paper assignment at the conclusion of class was intended to find out what they understood and where they had trouble.

- **Time Allotted for the Activity: 5 Minutes**

- **Details of the Implementation:**

- A "one-minute paper" is a very relatively short in-class writing assignment in which students respond to a question presented by the teacher in order to reflect on the day's lecture and give the teacher helpful feedback.
- At the end of the class, the teacher asked the following questions and the students wrote answers within a minute shortly which is shown in Fig. 1.
 - What is the concept taught in today's class that stands out in your mind?
 - What interesting questions remain unanswered about today's topic?
 - What relationship did you see between today's topic and other topics previously covered in this course?
- Based on the student's response in the sheet the topic which is not clear was addressed in the next session.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	3	2	2	1	2	1	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	1	1	2	1

Justification for correlation	The students apply the knowledge of optimization concepts in basic block.	The students analyze the block and identify optimization methods can be applied.	The students find the solutions for the given optimization problem.	The students responded individually which improve their skills.	The students apply the optimization concepts in developing software
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• **Images / Screenshot of the practice:**

1) Peephole Optimization, optimization of basic blocks, Representation of array, DAG construction

2) Understood everything.

3) Three address code - previous class
Assembly code - this class.

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K. Sairam

1, Peephole Optimization is clear,
DAG construction somewhat unclear.

2 Disjoint Acyclic Graph.

Today's class.

1. DAG - example.
 $t1 = i * 4$

2. Peephole optimization

Relation:-

Basic Block optimization
next step to optimization
from 3-address code to
assembly code.

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1. Dag, peepoptimization using Assembly code.

2. In previous topic result using three address code.

In peepoptimization result using Assembly code

1) peephole optimization
 2) peephole optimization.
 3) Yes, previous we saw 3-address code optimization, now we saw a part optimization of particular part of a target code, both leads to a strength-reduced & efficient code.

B. Dhasini

Today's Topic :-

- 1) DAG
- 2) Peephole Optimization.

~~Assembly~~

Previous topic result is 3 address code

Here it is assembly code

Fig 1: Sample Minute Paper

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Based on the students' response, we can identify that the students have a clear idea about the peephole optimization and DAG.
- Some of the students had doubt on the optimization using DAG.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The topics which is not clear to the students were identified and more explanation given to them which improve their understanding level and improved IAT marks in this topic.

❖ **Challenges faced in implementation:**

- To collect the view from all the students is a challenging task. Few of them not done the task and didn't submit the papers.

References:

- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/GSP_one-minute-paper.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/GAP_One_minute_Paper.pdf

Signature of Faculty Member

HOD